VANISHING AND COMPARISON STATEMENTS FOR THE COHOMOLOGY OF A PULLBACK OF A UNIFORMLY ANNIHILATED MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The notion of an (A, n, c)-module and the invariants q_0 = q(S) and d = d(S) of the graded situation. Set up in the author's section on a bit of uniformity and recalled in the statement itself. Fix the hypothesis on the module and the thresholds appearing in the five cases.
- The behaviour of the class of (A, n, c)-modules under short exact sequences. Invoked inside the counted proof at source line 5176 and attributed to a lemma of another chapter (a cross-chapter pointer). Allows the proof to replace the given module by a resolution by free modules over the quotient and keep control of the constants.

Source reference. Original lemma lemma-almost-exactness: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5190–5190. Navigation pointer only: the counted text refers to the Stacks item lemma-almost-exactness, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-annihilated: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5241–5241. Navigation pointer only: the counted text refers to the Stacks item lemma-annihilated, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-bound-q-and-d: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5200–5200. Navigation pointer

Date: stacks-project:local-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

only: the counted text refers to the Stacks item lemma-bound-q-and-d, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-ses-near-projective: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 5179–5179. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-ses-near-projective, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *With $q_0 = q(S)$ and $d = d(S)$ as above, suppose we have integers $n \geq c \geq 0$, an (A, n, c)-module M, an index $i \in \{0, 1, \ldots, d\}$, and an integer q. Then we distinguish the following cases*

1. *In the case $i = d \geq 1$ and $q \geq q_0$ we have $H^d(X, p^*M(q)) = 0$.*
2. *In the case $i = d - 1 \geq 1$ and $q \geq q_0$ we have $H^{d-1}(X, p^*M(q)) = 0$.*
3. *In the case $d-1 > i > 0$ and $q \geq q_0+(d-1-i)c$ the map $H^i(X, p^*M(q)) \to H^i(X, p^*M(q - (d - 1 - i)c))$ is zero.*
4. *In the case $i = 0$, $d \in \{0, 1\}$, and $q \geq q_0$, there is a surjection*
$$I^q M \longrightarrow H^0(X, p^*M(q))$$
5. *In the case $i = 0$, $d > 1$, and $q \geq q_0 + (d - 1)c$ the map*
$$H^0(X, p^*M(q)) \to H^0(X, p^*M(q - (d - 1)c))$$
has image contained in the image of the canonical map $I^{q-(d-1)c}M \to H^0(X, p^*M(q - (d - 1)c))$.

Proof. Let M be an (A, n, c)-module. Choose a short exact sequence
$$0 \to K \to (A/I^n)^{\oplus r} \to M \to 0$$
We will use below that K is an (A, n, c)-module, see More on Algebra, Lemma more-algebra-lemma-ses-near-projective. Consider the corresponding exact sequence
$$p^*K \to (\mathcal{O}_{Y_n})^{\oplus r} \to p^*M \to 0$$
We split this into short exact sequences
$$0 \to \mathcal{F} \to p^*K \to \mathcal{G} \to 0 \quad \text{and} \quad 0 \to \mathcal{G} \to (\mathcal{O}_{Y_n})^{\oplus r} \to p^*M \to 0$$
By Lemma lemma-almost-exactness the coherent module $\mathcal{F}$ is scheme theoretically supported on Y_c.

Proof of (1). Assume $d > 0$. We have to prove $H^d(X, p^*M(q)) = 0$ for $q \geq q_0$. By the vanishing of the cohomology of twists of $\mathcal{G}$ in degrees $> d$ and the long exact cohomology sequence associated to the second short exact sequence above, it suffices to prove that $H^d(X, \mathcal{O}_{Y_n}(q)) = 0$. This is true by Lemma lemma-bound-q-and-d.

Proof of (2). Assume $d > 1$. We have to prove $H^{d-1}(X, p^*M(q)) = 0$ for $q \geq q_0$. Arguing as in the previous paragraph, we see that it suffices to show that $H^d(X, \mathcal{G}(q)) = 0$. Using the first short exact sequence and the vanishing of the cohomology of twists of $\mathcal{F}$ in degrees $> d$ we see that it suffices to show $H^d(X, p^*K(q))$ is zero which is true by (1) and the fact that K is an (A, n, c)-module (see above).

Proof of (3). Let $0 < i < d-1$ and assume the statement holds for $i+1$ except in the case $i = d-2$ we have statement (2). Using the long exact sequence of cohomology associated to the second short exact sequence above we find an injection
$$H^i(X, p^*M(q - (d - 1 - i)c)) \subset H^{i+1}(X, \mathcal{G}(q - (d - 1 - i)c))$$

as $q-(d-1-i)c \geq q_0$ gives the vanishing of $H^i(X, \mathcal{O}_{Y_n}(q-(d-1-i)c))$ (see above). Thus it suffices to show that the map $H^{i+1}(X, \mathcal{G}(q)) \to H^{i+1}(X, \mathcal{G}(q-(d-1-i)c))$ is zero. To study this, we consider the maps of exact sequences

$$\begin{array}{ccccc}
H^{i+1}(X, p^*K(q)) & \longrightarrow & H^{i+1}(X, \mathcal{G}(q)) & \longrightarrow & H^{i+2}(X, \mathcal{F}(q)) \\
\downarrow & \swarrow & \downarrow & & \downarrow \\
H^{i+1}(X, p^*K(q-c)) & \longrightarrow & H^{i+1}(X, \mathcal{G}(q-c)) & \longrightarrow & H^{i+2}(X, \mathcal{F}(q-c)) \\
\downarrow & & \downarrow & & \\
H^{i+1}(X, p^*K(q-(d-1-i)c)) & \longrightarrow & H^{i+1}(X, \mathcal{G}(q-(d-1-i)c)) & &
\end{array}$$

Since $\mathcal{F}$ is scheme theoretically supported on Y_c we see that the canonical map $\mathcal{G}(q) \to \mathcal{G}(q-c)$ factors through $p^*K(q-c)$ by Lemma lemma-annihilated. This gives the dotted arrow in the diagram. (In fact, for the proof it suffices to observe that the vertical arrow on the extreme right is zero in order to get the dotted arrow as a map of sets.) Thus it suffices to show that $H^{i+1}(X, p^*K(q-c)) \to H^{i+1}(X, p^*K(q-(d-1-i)c))$ is zero. If $i = d-2$, then the source of this arrow is zero by (2) as $q - c \geq q_0$ and K is an (A, n, c)-module. If $i < d-2$, then as K is an (A, n, c)-module, we get from the induction hypothesis that the map is indeed zero since $q - c - (q-(d-1-i)c) = (d-2-i)c = (d-1-(i+1))c$ and since $q - c \geq q_0 + (d-1-(i+1))c$. In this way we conclude the proof of (3).

Proof of (4). Assume $d \in \{0, 1\}$ and $q \geq q_0$. Then the first short exact sequence gives a surjection $H^1(X, p^*K(q)) \to H^1(X, \mathcal{G}(q))$ and the source of this arrow is zero by case (1). Hence for all $q \in \mathbf{Z}$ we see that the map

$$H^0(X, (\mathcal{O}_{Y_n})^{\oplus r}(q)) \longrightarrow H^0(X, p^*M(q))$$

is surjective. For $q \geq q_0$ the source is equal to $(I^q/I^{q+n})^{\oplus r}$ by Lemma lemma-bound-q-and-d and this easily proves the statement.

Proof of (5). Assume $d > 1$. Arguing as in the proof of (4) we see that it suffices to show that the image of

$$H^0(X, p^*M(q)) \longrightarrow H^0(X, p^*M(q-(d-1)c))$$

is contained in the image of

$$H^0(X, (\mathcal{O}_{Y_n})^{\oplus r}(q-(d-1)c)) \longrightarrow H^0(X, p^*M(q-(d-1)c))$$

To show the inclusion above, it suffices to show that for $\sigma \in H^0(X, p^*M(q))$ with boundary $\xi \in H^1(X, \mathcal{G}(q))$ the image of ξ in $H^1(X, \mathcal{G}(q-(d-1)c))$ is zero. This follows by the exact same arguments as in the proof of (3). $\square$